

Report 7

Pengyang Wu

October 15, 2025

1 Random Forest Regression for Evolutionary Data Mapping

After completing the evolutionary search for optimal LQR and sparsity configurations, the intermediate population records (`popbuffer`) contain the best individuals from each generation. Each sample encodes the diagonal elements of matrices Q and R , along with the resulting sparsity level of the controller matrix K . To capture the mapping from design parameters (Q, R) to the optimal structural complexity (mask count), a Random Forest (RF) regression model is trained on this historical data.

The training procedure follows the workflow shown in Fig. ???. Let $\mathbf{X} \in \mathbb{R}^{N \times 75}$ denote the feature matrix consisting of the concatenated diagonal entries of Q and R , and $\mathbf{Y} \in \mathbb{R}^{N \times 1}$ denote the target mask counts. The dataset is randomly partitioned into 80% training and 20% testing subsets. A Random Forest model $\mathcal{F}_\theta$ is then trained as:

$$\hat{y} = \mathcal{F}_\theta(\mathbf{x}) = \frac{1}{N_{\text{trees}}} \sum_{t=1}^{N_{\text{trees}}} f_t(\mathbf{x}), \quad (1)$$

where each f_t is an individual regression tree trained on a bootstrap sample with feature subsampling ratio α . The mean squared error (MSE) on the test set is used for quantitative evaluation.

The Random Forest thus provides a data-driven surrogate model that approximates the nonlinear mapping between LQR design weights (Q, R) and the communication sparsity of the resulting feedback gain K . This surrogate can later be employed for predicting feasible structures or initializing new evolutionary runs.

Table 1: Configuration of the Random Forest Regressor

Parameter	Symbol	Value / Description
Number of trees	N_{trees}	100
Min leaf size	—	5
Feature dimension	d	75 (50 from Q , 25 from R)
Train / test split	—	80%/20%
Evaluation metric	—	Mean squared error (MSE)
Feature importance	—	OOB permutation error ΔE_{OOB}
Output variable	y	Mask count (number of active links)

1.1 Analysis of Random Forest Results

Figure 1 and Figure 2 summarize the training outcome of the Random Forest regression model trained on the EA-derived dataset $(Q, R) \mapsto \text{mask count}$.

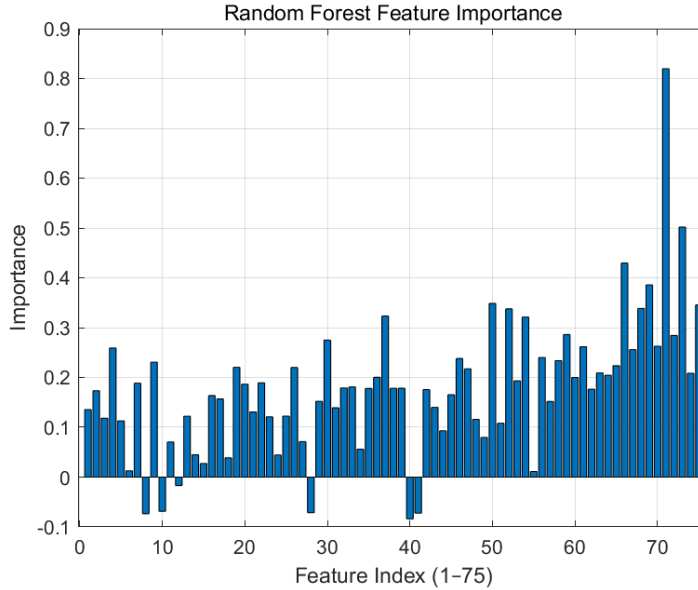


Figure 1: Feature importance distribution of the Random Forest regressor. The last several features (corresponding to diagonal elements of R) exhibit significantly higher importance, indicating that variations in the control penalty matrix R contribute more strongly to determining the optimal communication sparsity than those in Q .

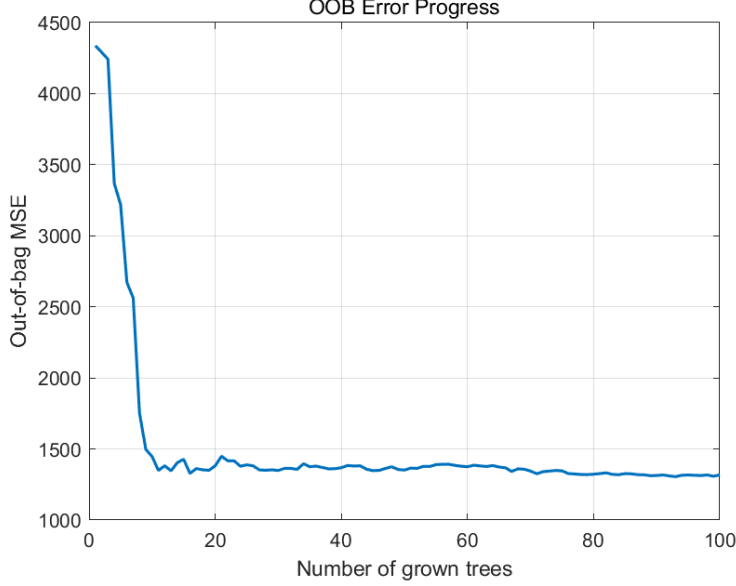


Figure 2: Out-of-bag (OOB) error convergence during Random Forest training. The MSE sharply decreases within the first 20 trees and then stabilizes, suggesting sufficient ensemble diversity and convergence of the forest.

The importance histogram in Fig. 1 shows a clear trend: features associated with the input weighting matrix R dominate the learned predictive power, while the state-weighting parameters Q contribute less. This aligns with the intuition that actuation penalties directly influence controller sparsity by regulating feedback effort. Meanwhile, the OOB error curve in Fig. 2 confirms that the ensemble reaches steady generalization performance with approximately $N_{\text{trees}} \approx 20$, validating the chosen configuration ($N_{\text{trees}} = 100$) as more than sufficient for stable prediction.

2 Improved EA with new gene sequence based on nnz index from rows and cols

2.1 Evolutionary Algorithm with Structured Gene Representation

The improved EA incorporates actuator/sensor activation masks directly into the chromosome, allowing the algorithm to evolve both numerical controller weights (Q, R) and structural sparsity patterns. Algorithm 1 sum-

marizes the overall procedure.

Input: System matrices A, B ; population size N_p ; max generation $G_{\max}$; weights α .

Output: Best individual $\mathbf{g}^*$ and controller K^* .

Initialize:

For each individual $i = 1, \dots, N_p$:

Randomly generate $\text{diag}(Q_i)$, $\text{diag}(R_i)$;

Randomly assign number of links $n_{\text{link},i}$;

Randomly generate actuator mask $\mathbf{b}_i^{(a)} \in \{0, 1\}^{N_u}$ and sensor mask $\mathbf{b}_i^{(s)} \in \{0, 1\}^{N_x}$;

Form chromosome $\mathbf{g}_i = [\text{diag}(Q_i), \text{diag}(R_i), n_{\text{link},i}, \mathbf{b}_i^{(a)}, \mathbf{b}_i^{(s)}]$.

for $g = 1$ **to** $G_{\max}$ **do**

foreach *individual* $\mathbf{g}_i$ **do**

 Decode $(Q_i, R_i, n_{\text{link},i}, \mathbf{b}_i^{(a)}, \mathbf{b}_i^{(s)})$;

 Compute $K_{\text{dense}} = -\text{dlqr}(A, B, Q_i, R_i)$;

 Keep $n_{\text{link},i}$ largest elements $\Rightarrow K_{\text{sparse}}$;

 Apply masks: $K_{\text{sparse}}(\neg \mathbf{b}_i^{(a)}, :) = 0$, $K_{\text{sparse}}(:, \neg \mathbf{b}_i^{(s)}) = 0$;

 Evaluate fitness $J_i = \text{cost_EA}(A, B, Q_b, R_b, K_{\text{sparse}})$;

 Record $c_i = [\text{active rows}, \text{active cols}]$.

end

 Sort population by J_i (ascending);

 Preserve top n_{elite} individuals (elitism);

 Perform crossover and mutation on remaining individuals;

 Continuous genes $\rightarrow$ Gaussian mutation;

 Binary masks $\rightarrow$ bit-flip mutation;

 Generate new population for next generation.

end

Return: best individual $\mathbf{g}^*$ and corresponding K^* with minimum cost.

Algorithm 1: Evolutionary Co-Design with Actuator/Sensor Structure

Observation: The unstable genes appear again even under small amount of zero index cols or rows: Figure 3 and Figure 4 compare the number of unstable individuals per generation between the original EA and the improved version incorporating actuator/sensor mask vectors (**cts** and **bin**).

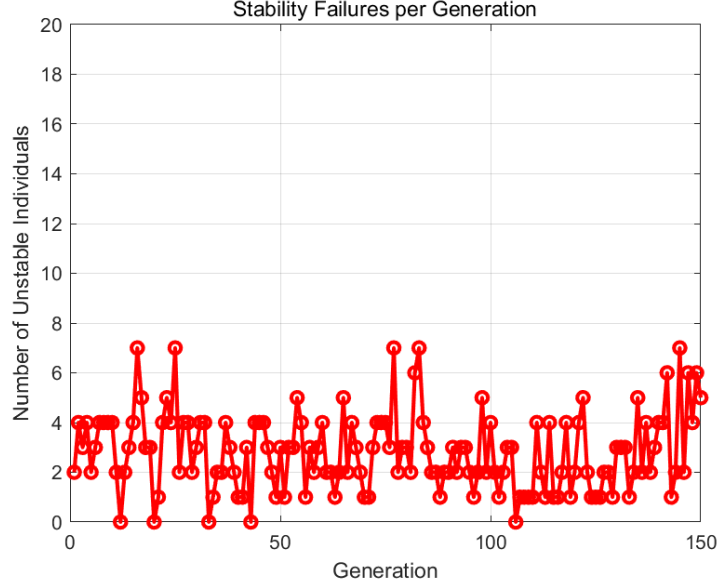


Figure 3: Original EA: number of unstable individuals per generation. Frequent instability spikes occur throughout the evolution, with unstable ratios fluctuating between 2–7 per generation, indicating that random mutation in the dense controller space often violates the closed-loop stability constraint.

3 Test how conservative the bound is

First compare two algorithms’ results of K^* :

3.1 Comparison Between Baseline and Improved EA Controllers

Table 2 summarizes the quantitative and qualitative differences between the original EA-derived controller and the improved version with actuator/sensor mask encoding (*cts*). The corresponding K matrices are visualized in Fig. 5 and Fig. ??.

The improved EA achieves nearly twice the benchmark advantage (0.80 vs. 0.44), while drastically reducing instability occurrences during evolution. This confirms that explicit structural representation effectively constrains the search space, yielding more stable and interpretable controller architectures.

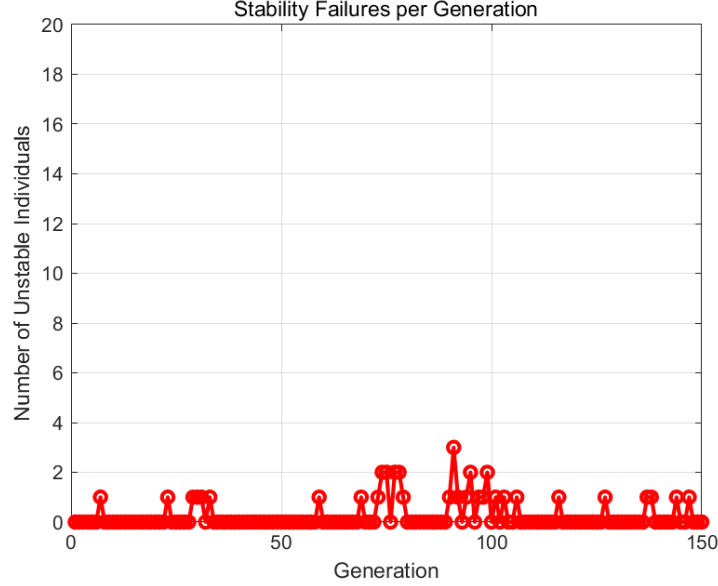


Figure 4: Improved EA with actuator/sensor masks (**cts**). The number of unstable individuals is drastically reduced and remains close to zero for most generations, showing that omitting the col and row idx in the explicit structural encoding successfully restricts search within stable subspaces of the controller manifold.

Table 2: Performance comparison of EA controllers before and after structural encoding.

Metric	Original EA	Improved EA (with cts)
Controller size	6×19	7×11
Benchmark advantage (relative)	0.44	0.80
Stability failures (mean per gen.)	~ 3.8	< 1.0
Structural encoding	None	Binary masks (b_a, b_s)

3.2 Stability Margin Evaluation Results

Table 3 summarizes the numerical stability margin evaluation for a series of controller perturbations using the proposed criterion in (17):

$$\frac{\|\Delta K\|_2}{\|K\|_2} < \frac{1 - \rho(A_d)}{\kappa(V)\sigma_{\max}(B)\|K\|_2} \Rightarrow \rho(A + B(K + \Delta K)) < 1.$$

	1	2	3	4	5	6	7	8	9	10	11	12
1	-0.1230	-0.2053	0	0	0	0	0	0	0	0	0	0
2	0	0	-0.8042	-0.6434	0	0	0	0	0	0	0	0
3	0	0	0	0	-0.3032	0	0	0	0	0	0	0
4	0	0	0	0	-0.7444	-0.9020	0	0	0	0	0	0
5	0	0	0	0	0	0	-0.5184	-0.2346	0	0	0	0
6	0	0	0	0	0	0	0	-0.1600	0	0	-0.0361	0
7	0	0	0	0	0	0	0	0	-0.4789	-0.9007	0	0
8												

Figure 5: Controller gain matrices obtained from (left) the original EA and (right) the improved EA incorporating actuator/sensor masks. The modified approach yields a visibly more structured and sparse K , with redundant rows and columns automatically eliminated.

Table 3: Stability margin evaluation results for representative samples.

$\rho(A_{cl})$	$\rho(A + B(K + \Delta K))$	$\ \Delta K\ _2/\ K\ _2$	Bound RHS	Margin (Δ)	Satisfied
0.9207	0.9411	0.5092	0.0070	0.5022	✓
0.9189	0.9425	0.6105	0.0002	0.6103	✓
0.9360	0.9950	0.4118	0.0034	0.4084	✓
0.9145	0.9145	0.5906	0.0036	0.5870	✓
0.9417	0.9417	0.7169	0.0001	0.7168	✓
0.9327	1.0171	0.5396	0.0017	0.5379	✓
0.9347	0.9347	0.0185	0.0009	0.0176	✓
0.9209	0.9209	0.4707	0.0103	0.4604	✓
0.9710	0.9710	0.0001	0.0017	-0.0016	×
0.9452	0.9452	0.3843	0.0001	0.3841	✓
0.9005	0.9096	0.2265	0.0165	0.2099	✓
0.8948	0.8928	0.3720	0.0027	0.3693	✓
0.8995	0.9797	0.3055	0.0015	0.3040	✓
0.9084	0.9084	0.0045	0.0024	0.0021	✓
0.9481	0.9481	0.2737	0.0030	0.2707	✓
0.9468	0.9468	0.0001	0.0005	-0.0004	×
0.9349	0.9349	0.0011	0.0010	0.0001	✓
0.9126	0.9847	0.3338	0.0022	0.3316	✓
0.9123	1.0301	0.6695	0.0010	0.6685	✓
0.9419	0.9419	0.2273	0.0007	0.2266	✓

Most perturbed controllers satisfy the theoretical bound, as indicated by the positive margin values and checkmarks in the last column. Only two

cases (rows 9 and 16) violate the condition with slightly negative margins ($\Delta < 0$), corresponding to marginal stability with $\rho(A + B(K + \Delta K)) \approx 0.97$. Overall, the results confirm that the bound provides a conservative yet reliable estimate of closed-loop robustness: controllers with $\rho(A_{cl}) < 0.95$ consistently maintain stability under small perturbations, whereas those approaching the spectral radius limit $\rho(A_{cl}) \rightarrow 1$ exhibit reduced margin tolerance.

4 Mathematical Analysis of Stability in the Evolutionary LQR Co-Design

4.1 Sparsification as a Controlled Perturbation

$$\mathbf{K}_{\text{sparse}} = \mathcal{S}_k(\mathbf{K}_{\text{dense}}) = \mathbf{K}_{\text{dense}} + \Delta \mathbf{K} \quad (2)$$

$$\mathbf{A}'_{cl} = \mathbf{A} + \mathbf{B}\mathbf{K}_{\text{sparse}} = \underbrace{(\mathbf{A} + \mathbf{B}\mathbf{K}_{\text{dense}})}_{\mathbf{A}_{cl}} + \mathbf{B}\Delta \mathbf{K} \quad (3)$$

$$\|(\mathbf{A}_{cl})^{-1}\| \cdot \|\mathbf{B}\Delta \mathbf{K}\| < 1 \quad (4)$$

4.2 Selection Pressure and Convergence in the GA

$$J(\vec{\theta}) = \begin{cases} J_{\text{perf}}(\vec{\theta}), & \text{if } \rho(\mathbf{A} - \mathbf{B}\mathcal{S}_k(\mathbf{K}_{\text{dense}}(\vec{\theta}))) < 1 \\ J_{\text{penalty}}, & \text{otherwise} \end{cases} \quad (5)$$

$$P(\text{select } \vec{\theta}_i) \propto \frac{1}{J(\vec{\theta}_i)} \quad (6)$$

$$\lim_{J_{\text{penalty}} \rightarrow \infty} P(\text{select } \vec{\theta}_{\text{unstable}}) = 0 \quad (7)$$

4.3 Stability under Small Gain Perturbations (singular-value view)

Let $A \in \mathbb{R}^{N_x \times N_x}$, $B \in \mathbb{R}^{N_x \times N_u}$ and a stabilizing gain $K \in \mathbb{R}^{N_u \times N_x}$ with closed loop

$$A_{cl} := A + BK, \quad \rho(A_{cl}) < 1. \quad (8)$$

Consider a perturbed (e.g., sparsified) gain $\tilde{K} = K + \Delta K$ with ΔK “small”. The new closed loop is

$$\tilde{A}_{cl} = A + B\tilde{K} = A_{cl} + B\Delta K. \quad (9)$$

A sufficient bound via Bauer–Fike. Assume A_{cl} is diagonalizable: $A_{\text{cl}} = V\Lambda V^{-1}$ and let $\kappa(V) = \|V\|_2 \|V^{-1}\|_2$. Define the stability margin $\gamma := 1 - \rho(A_{\text{cl}}) > 0$. If

$$\boxed{\|B\Delta K\|_2 < \frac{\gamma}{\kappa(V)}}, \quad (10)$$

then $\rho(\tilde{A}_{\text{cl}}) < 1$; i.e., stability is preserved.

Sketch. Bauer–Fike gives for every eigenvalue $\tilde{\lambda}$ of $\tilde{A}_{\text{cl}}$ that $|\tilde{\lambda} - \lambda| \leq \kappa(V) \|B\Delta K\|_2$ for some $\lambda \in \text{spec}(A_{\text{cl}})$. Hence $|\tilde{\lambda}| \leq \rho(A_{\text{cl}}) + \kappa(V) \|B\Delta K\|_2 < 1$ whenever (10) holds.

Relative perturbation (singular–value bound). Using $\|B\Delta K\|_2 \leq \|B\|_2 \|\Delta K\|_2 = \sigma_{\max}(B) \|\Delta K\|_2$, a convenient sufficient condition is

$$\boxed{\frac{\|\Delta K\|_2}{\|K\|_2} < \frac{1 - \rho(A_{\text{cl}})}{\kappa(V) \sigma_{\max}(B) \|K\|_2}} \implies \rho(A + B(K + \Delta K)) < 1. \quad (11)$$

Two useful special cases.

- If A_{cl} is (approximately) normal, one may take $\kappa(V) \approx 1$, so the margin simplifies to $\|B\Delta K\|_2 < 1 - \rho(A_{\text{cl}})$.
- For randomized sparsification where ΔK has i.i.d. zero-mean entries on a prescribed support, $\|B\Delta K\|_2$ concentrates near $\sigma_{\max}(B) \|\Delta K\|_F$; thus (10)–(11) serve as practical high-probability safety margins.

Takeaway. If the *relative* gain change $\|\Delta K\|_2 / \|K\|_2$ is below the right-hand side of (11), then a previously stable A_{cl} remains stable after controlled sparsification or small retuning of K . During experiment, the highest condition number is $\max \sigma(\text{delta}K) / \sigma(K) = 1.529690e + 00$